

Equity Valuation Casebook for MBA Students

Relative Valuation

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RELATIVE VALUATION – TWO STAGE MODEL PROBLEMS

Case 35A: Two-Stage P/E Valuation — Specialty Retailer

Learning Objective: Estimate a justified P/E for a company with temporary high growth and changing payout.

Case Situation

An Indian specialty retailer currently has an EPS of ₹10.00 and is expected to grow earnings at 25% annually for the next five years while paying out only 10% of earnings during this expansion. Thereafter, growth is expected to stabilize at 8%, supported by a sustainable ROE of 18%, with payout rising to a level consistent with that stable growth. The cost of equity is 14%. The stock currently trades at 20x trailing earnings.

Required

1. Estimate the EPS and dividend for each of the five high-growth years.
2. Estimate the stable-stage payout, terminal value, and the implied two-stage justified P/E.
3. Assess whether the prevailing market multiple of 20x is justified.

Solution

A. Relevant Valuation Principle: The complete two-stage P/E has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal value based on stable-stage payout, growth, and ROE. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — EPS, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4	Year 5
EPS (EPS_t)	$EPS_{(t-1)} \times (1+g_{hg})$	12.50	15.63	19.53	24.41	30.52
Dividend (D_t)	$EPS_t \times \text{payout}_{hg}$	1.25	1.56	1.95	2.44	3.05
PV of Dividend	$D_t \div (1+K_e)^t$	1.10	1.20	1.32	1.45	1.58

Sum of PV of high-growth-phase dividends = ₹6.65

C. Stable Growth Period

Step 2 — Stable-stage payout ratio

Formula: $\text{payout_stable} = 1 - (g_stable \div \text{ROE_stable})$

Substitution: $= 1 - (0.08 \div 0.18)$

Result: **payout_stable = 55.56%**

Step 3 — EPS in the first stable year (Year 6)

Formula: $\text{EPS6} = \text{EPS5} \times (1 + g_stable)$

Substitution: $= 30.52 \times 1.08$

Result: **EPS6 = ₹32.96**

Step 4 — Dividend in the first stable year

Formula: $D6 = \text{EPS6} \times \text{payout_stable}$

Substitution: $= 32.96 \times 0.5556$

Result: **D6 = ₹18.31**

Step 5 — Terminal value at the end of Year 5

Formula: $\text{TV5} = D6 \div (K_e - g_stable)$

Substitution: $= 18.31 \div (0.14 - 0.08)$

Result: **TV5 = ₹305.18**

Step 6 — Present value of TV5

Formula: $\text{PV}(\text{TV5}) = \text{TV5} \div (1 + K_e)^5$

Substitution: $= 305.18 \div (1.14)^5$

Result: **PV(TV5) = ₹158.50**

D. Combining the Two Components

Step 7 — Total value today (V0)

Formula: $V0 = \text{Sum of PV}(D_t) + \text{PV}(\text{TV5})$

Substitution: $= 6.65 + 158.50$

Result: **V0 = ₹165.15**

Step 8 — Implied two-stage justified P/E

Formula: $\text{Implied P/E} = V0 \div \text{EPS0}$

Substitution: $= 165.15 \div 10.00$

Result: **Implied P/E = 16.51x**

E. Final Result: Implied two-stage justified P/E $\approx 16.51x$, against a market P/E of 20x.

F. Interpretation: The market multiple of 20x sits meaningfully above the fundamentals-based justified multiple of 16.51x, suggesting the current price embeds either a longer or steeper high-growth period

than modeled here, or a lower perceived risk — the retailer's aggressive 25% growth assumption already looks generous, so the market appears to be pricing in an even more optimistic scenario.

G. Decision/Recommendation: Given the market multiple already exceeds a fairly optimistic justified P/E, the analyst should treat the stock as, at best, fully valued, and should avoid recommending fresh purchases at the prevailing 20x multiple without a more aggressive (and less easily defensible) growth assumption.

H. Teaching Insight: This case shows that even a generous high-growth assumption (25% for five years) does not always justify the prevailing market multiple — students should resist the temptation to keep raising growth or extending its duration merely to make the model match the market price.

Case 35B: Two-Stage P/E Valuation — Renewable Energy (Solar) Company

Learning Objective: Estimate a justified P/E when the high-growth phase is short but exceptionally rapid, and observe how quickly value becomes almost entirely dependent on terminal-stage assumptions when both the growth window is brief and payout is minimal.

Case Situation

Suryodaya Renewable Power Ltd, an Indian solar power developer, currently has an EPS of ₹8.00 and is expected to grow earnings at 35% annually for the next three years as it commissions a pipeline of new solar capacity, paying out only 5% of earnings during this phase as nearly all cash is deployed into new projects. Thereafter, growth is expected to stabilize at 7%, supported by a sustainable ROE of 15% once the portfolio matures into steady, utility-like cash flows. The cost of equity is 15%, reflecting the sector's project execution and regulatory risks. The stock currently trades at 18x trailing earnings.

Required

1. Estimate the EPS and dividend for each of the three high-growth years.
2. Estimate the stable-stage payout, terminal value, and the implied two-stage justified P/E.
3. Assess whether the prevailing market multiple of 18x is justified.

Solution

A. Relevant Valuation Principle: The complete two-stage P/E has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal value based on stable-stage payout, growth, and ROE. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — EPS, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3
EPS (EPS_t)	$EPS_{(t-1)} \times (1+g_{hg})$	10.80	14.58	19.68
Dividend (D_t)	$EPS_t \times \text{payout}_{hg}$	0.54	0.73	0.98
PV of Dividend	$D_t \div (1+K_e)^t$	0.47	0.55	0.65

Sum of PV of high-growth-phase dividends = ₹1.67

C. Stable Growth Period

Step 2 — Stable-stage payout ratio

Formula: $\text{payout_stable} = 1 - (g_{\text{stable}} \div \text{ROE_stable})$

Substitution: $= 1 - (0.07 \div 0.15)$

Result: **payout_stable = 53.33%**

Step 3 — EPS in the first stable year (Year 4)

Formula: $EPS_4 = EPS_3 \times (1+g_{\text{stable}})$

Substitution: $= 19.68 \times 1.07$

Result: **EPS4 = ₹21.06**

Step 4 — Dividend in the first stable year

Formula: $D_4 = EPS_4 \times \text{payout_stable}$

Substitution: $= 21.06 \times 0.5333$

Result: **D4 = ₹11.23**

Step 5 — Terminal value at the end of Year 3

Formula: $TV_3 = D_4 \div (K_e - g_{\text{stable}})$

Substitution: $= 11.23 \div (0.15 - 0.07)$

Result: **TV3 = ₹140.41**

Step 6 — Present value of TV3

Formula: $PV(TV_3) = TV_3 \div (1+K_e)^3$

Substitution: $= 140.41 \div (1.15)^3$

Result: **PV(TV3) = ₹92.32**

D. Combining the Two Components

Step 7 — Total value today (V0)

Formula: $V_0 = \text{Sum of } PV(D_t) + PV(TV_3)$

Substitution: $= 1.67 + 92.32$

Result: **V0 = ₹93.99**

Step 8 — Implied two-stage justified P/E

Formula: $\text{Implied P/E} = V_0 \div \text{EPS}_0$

Substitution: $= 93.99 \div 8.00$

Result: **Implied P/E = 11.75x**

E. Final Result: Implied two-stage justified P/E $\approx 11.75x$, against a market P/E of 18x.

F. Interpretation: With only three years of high growth at a minimal 5% payout, the terminal value carries over 98% of total worth, which is even more extreme than Case 35A's already-terminal-heavy 96%. The market's 18x multiple is roughly 35% above the justified figure, a wider gap than 35A's — for a young, capex-intensive sector, this signals the market may be pricing in either a longer runway of 35% growth than three years, faster-than-modeled capacity additions, or is discounting execution and regulatory risk less heavily than the 15% K_e assumed here.

G. Decision/Recommendation: Given the market multiple sits well above a justified figure that is itself sensitive to just a handful of terminal-stage assumptions, the analyst should treat 18x as pricing in considerable optimism and should stress-test the stable-stage ROE and growth assumptions specifically — in a sector this reliant on terminal value, small changes in those two inputs alone can close much of the gap.

H. Teaching Insight: This case sharpens the terminal-value-dominance lesson from Case 35A by shortening the explicit window from five years to three and thinning the payout further — students should notice that a shorter high-growth phase does not mean less exposure to terminal-value risk; if anything, it means the analysis rests on the stable-stage assumptions even more heavily, that much sooner.

Case 35C: Two-Stage P/E Valuation — IT Services Company

Learning Objective: Estimate a justified P/E for a steady, longer-duration grower that pays a meaningful dividend even during its growth phase and see how a real payout sustained over a longer explicit window shifts the balance of value away from the terminal stage.

Case Situation

Vistaar Technologies Ltd, an Indian IT services company, currently has an EPS of ₹15.00 and is expected to grow earnings at a steadier 15% annually for the next six years as it scales its digital-services practice, paying out a meaningful 30% of earnings during this phase given the business's asset-light, cash-generative nature. Thereafter, growth is expected to stabilize at 6%, supported by a sustainable ROE of 20%, with payout rising further to a level consistent with that stable growth. The cost of equity is 13%, reflecting the company's scale and earnings predictability. The stock currently trades at 22x trailing earnings.

Required

1. Estimate the EPS and dividend for each of the six high-growth years.
2. Estimate the stable-stage payout, terminal value, and the implied two-stage justified P/E.
3. Assess whether the prevailing market multiple of 22x is justified.

Solution

A. Relevant Valuation Principle: The complete two-stage P/E has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal value based on stable-stage payout, growth, and ROE. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — EPS, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
EPS (EPS_t)	$EPS_{t-1} \times (1+g_{hg})$	17.25	19.84	22.81	26.24	30.17	34.70
Dividend (D_t)	$EPS_t \times \text{payout}_{hg}$	5.17	5.95	6.84	7.87	9.05	10.41
PV of Dividend	$D_t \div (1+Ke)^t$	4.58	4.66	4.74	4.83	4.91	5.00

Sum of PV of high-growth-phase dividends = ₹28.72

C. Stable Growth Period

Step 2 — Stable-stage payout ratio

Formula: $\text{payout}_{stable} = 1 - (g_{stable} \div ROE_{stable})$

Substitution: $= 1 - (0.06 \div 0.20)$

Result: **payout_{stable} = 70.0%**

Step 3 — EPS in the first stable year (Year 7)

Formula: $EPS_7 = EPS_6 \times (1+g_{stable})$

Substitution: $= 34.70 \times 1.06$

Result: **EPS₇ = ₹36.78**

Step 4 — Dividend in the first stable year

Formula: $D_7 = EPS_7 \times \text{payout}_{stable}$

Substitution: $= 36.78 \times 0.70$

Result: **D₇ = ₹25.74**

*Step 5 — Terminal value at the end of Year 6*Formula: $TV_6 = D_7 \div (K_e - g_{\text{stable}})$ Substitution: $= 25.74 \div (0.13 - 0.06)$ Result: **TV6 = ₹367.78***Step 6 — Present value of TV6*Formula: $PV(TV_6) = TV_6 \div (1 + K_e)^6$ Substitution: $= 367.78 \div (1.13)^6$ Result: **PV(TV6) = ₹176.65****D. Combining the Two Components***Step 7 — Total value today (V0)*Formula: $V_0 = \text{Sum of } PV(D_t) + PV(TV_6)$ Substitution: $= 28.72 + 176.65$ Result: **V0 = ₹205.37***Step 8 — Implied two-stage justified P/E*Formula: $\text{Implied P/E} = V_0 \div \text{EPS}_0$ Substitution: $= 205.37 \div 15.00$ Result: **Implied P/E = 13.69x****E. Final Result:** Implied two-stage justified P/E $\approx 13.69x$, against a market P/E of 22x.

F. Interpretation: Here, the explicit high-growth phase contributes a much more substantial 14% of total value — a sharp contrast to 35A (4%) and 35B (under 2%) — because a longer six-year window combined with a real 30% payout means meaningful cash reaches shareholders well before the terminal stage. Even so, the market's 22x multiple sits about 38% above the justified 13.69x, a wider proportional gap than either 35A or 35B, despite this being the most "grounded" of the three cases in terms of how much of its value is verifiable in the near term.

G. Decision/Recommendation: Because a larger share of this valuation rests on the explicit, more forecastable six-year period rather than the terminal stage alone, the analyst has comparatively more confidence in the 13.69x figure than in 35A or 35B's justified multiples, which makes the 38% gap to the market price harder to explain away as terminal-assumption uncertainty. The stock should be treated as significantly overvalued at 22x, with the burden of proof on anyone arguing that growth will run meaningfully longer than six years.

H. Teaching Insight: Read alongside 35A and 35B, this case completes a three-way spectrum: a short, thin-payout, terminal-dominated story (35B, 98% terminal); a moderate story (35A, 96% terminal); and a longer, real-payout story where the explicit period finally contributes a meaningful share (35C, 86% terminal). Students should take from this progression that terminal-value dominance is not an inherent

flaw of the two-stage model — it is a direct, quantifiable consequence of how short the growth window is and how little of it is paid out along the way.

Case 42A: Two-Stage P/B Valuation — Private-Sector Lender (zero growth-phase payout)

Learning Objective: Value a company whose high ROE and growth are expected to decline to stable levels.

Case Situation

An Indian private-sector lender currently has a book value of ₹100 per share and an ROE of 22%, which is expected to persist for five years as the bank fully retains earnings to fund rapid loan-book growth (zero payout during this phase). Thereafter, ROE is expected to normalize to 15% with a stable payout ratio of 40%. The cost of equity is 14%. An analyst wants the two-stage justified P/B implied by this profile.

Required

1. Project the book value per share at the end of the five-year high-growth phase.
2. Estimate the present value of any dividends paid during the high-growth phase, and the terminal-stage justified P/B and value at year five.
3. Estimate the implied two-stage justified P/B as of today.

Solution

A. Relevant Valuation Principle: The complete two-stage P/B has two components: the present value of dividends paid during the high-growth phase, plus the present value of a stable-stage terminal multiple applied to book value at the end of that phase. Each year's dividend is computed on that year's own book value.

B. Formula: $BV_t = BV_0 \times (1 + g_{hg})^t$, where $g_{hg} = (1 - \text{payout}_{hg}) \times \text{ROE}_{hg}$; $D_t = \text{payout}_{hg} \times \text{ROE}_{hg} \times BV_t$; Stable-stage Justified P/B = $(\text{ROE}_{st} - g_{st}) / (K_e - g_{st})$; Value5 = $BV_5 \times \text{Justified P/B}$; $PO = \sum PV(D_t) + PV(\text{Value5})$

C. Substitution and Working:

Retention during growth phase = 100%; $g_{hg} = 1.00 \times 0.22 = 22.0\%$

BV (years 1–5): 122.00, 148.84, 181.58, 221.53, 270.27

Dividends, $D_t = 0.00 \times 0.22 \times BV_t$ (years 1–5): ₹0, 0, 0, 0, 0 (*payout_{hg} = 0%, so this line is zero regardless of which book value it's paired with*)

PV of high-growth-phase dividends = ₹0.00

Stable-stage: $b = 60\%$, $g = 0.60 \times 0.15 = 9.0\%$; Justified P/B (stable) = $(0.15 - 0.09) / (0.14 - 0.09) = 1.20x$

Value5 = $270.27 \times 1.20 = ₹324.32$; PV of Value5 = $324.32 \times (1.14)^{-5} = ₹168.44$

$PO = 0.00 + 168.44 = ₹168.44$; Implied two-stage justified P/B = $168.44 / 100 = 1.68x$

D. Final Result: Book value grows to ₹270.27 per share by year 5; implied two-stage justified P/B as of today $\approx 1.68x$.

E. Interpretation: With zero payout during the growth phase, the entire justified value comes from the terminal component. This is the one case of the three where the choice between same-year and lagged book value for the dividend calculation is irrelevant — because the dividend itself is zero either way.

F. Decision/Recommendation: The analyst can use 1.68x as the fundamentals-based justified P/B benchmark for the lender today, to be compared against its current market P/B multiple before forming a recommendation.

G. Teaching Insight: High current ROE alone does not translate one-for-one into a high present-day justified P/B — discounting and the eventual stable-stage multiple both matter.

Case 42B: Two-Stage P/B Valuation — Housing Finance Company (20% growth-phase payout)

Learning Objective: Value a company whose high ROE and growth decline to stable levels, where the high-growth phase itself pays a modest dividend computed on that year's own book value.

Case Situation

Griha Nirman Housing Finance Ltd, an Indian housing-finance company, currently has a book value of ₹150 per share and an ROE of 20%, expected to persist for four years as the company grows its mortgage book — but unlike a pure-retention growth story, the company maintains a modest 20% payout even during this phase to satisfy income-seeking shareholders. Thereafter, ROE is expected to normalize to 14% with a stable payout ratio of 50%. The cost of equity is 13%. An analyst wants the two-stage justified P/B implied by this profile.

Required

1. Project the book value per share at the end of the four-year high-growth phase, accounting for the 20% payout.
2. Estimate the present value of the dividends paid during the high-growth phase, and the terminal-stage justified P/B and value at year four.
3. Estimate the implied two-stage justified P/B as of today.

Solution

A. Relevant Valuation Principle: The complete two-stage P/B has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal multiple applied to book value at the end of that phase. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — Growth rate during the high-growth phase

Formula: $g_{hg} = (1 - \text{payout}_{hg}) \times \text{ROE}_{hg}$

Substitution: $g_{hg} = (1 - 0.20) \times 0.20$

Result: **$g_{hg} = 16.0\%$**

Step 2 — Book value, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4
Book Value (BV_t)	$BV_0 \times (1+g_{hg})^t$	174.00	201.84	234.13	271.60
Dividend (D_t)	$\text{payout}_{hg} \times ROE_{hg} \times BV_t$	6.96	8.07	9.37	10.86
PV of Dividend	$D_t \div (1+Ke)^t$	6.16	6.32	6.49	6.66

Sum of PV of high-growth-phase dividends = ₹25.64

C. Stable Growth Period

Step 3 — Stable-stage growth rate

Formula: $g_{st} = (1 - \text{payout}_{st}) \times ROE_{st}$

Substitution: $g_{st} = (1 - 0.50) \times 0.14$

Result: **$g_{st} = 7.0\%$**

Step 4 — Stable-stage justified P/B

Formula: $\text{Justified P/B(stable)} = (ROE_{st} - g_{st}) / (Ke - g_{st})$

Substitution: $= (0.14 - 0.07) / (0.13 - 0.07)$

Result: **Justified P/B(stable) = 1.17x**

Step 5 — Value at the end of the high-growth period (Year 4)

Formula: $\text{Value4} = BV_4 \times \text{Justified P/B(stable)}$

Substitution: $= 271.60 \times 1.17$

Result: **Value4 = ₹316.86**

Step 6 — Present value of Value4

Formula: $PV(\text{Value4}) = \text{Value4} \div (1+Ke)^4$

Substitution: $= 316.86 \div (1.13)^4$

Result: **PV(Value4) = ₹194.34**

D. Combining the Two Components

Step 7 — Total value today (P0)

Formula: $P0 = \text{Sum of } PV(D_t) + PV(\text{Value4})$

Substitution: $= 25.64 + 194.34$

Result: **P0 = ₹219.97**

Step 8 — Implied two-stage justified P/B

Formula: Implied P/B = $P_0 \div BV_0$

Substitution: $= 219.97 \div 150$

Result: **Implied P/B = 1.47x**

E. Final Result: Book value grows to ₹271.60 per share by year 4; the high-growth phase separately contributes ₹25.64 of present value through its dividends; implied two-stage justified P/B as of today $\approx$ 1.47x.

F. Interpretation: Because each dividend is paired with that year's own (already-grown) book value rather than the smaller prior-year figure, every dividend in the stream is slightly larger than under a lagged-book-value convention — the interim-dividend component comes to ₹25.64 (about 12% of total value), pulling the overall justified P/B up to 1.47x.

G. Decision/Recommendation: The analyst should use 1.47x, built from both the interim-dividend and terminal components on a consistent same-year book-value basis, as the benchmark.

H. Teaching Insight: Once a growth-phase payout is introduced, students must fix two things at once: include the interim-dividend term at all, and apply it consistently to the same year's book value throughout the projection — not a mix of current- and prior-year figures.

Case 42C: Two-Stage P/B Valuation — Small Finance Bank (10% growth-phase payout)

Learning Objective: Value a company with a very high growth-phase ROE and a modest payout over a longer six-year window, applying the dividend consistently to each year's own book value.

Case Situation

Uday Small Finance Bank Ltd, a recently listed Indian small finance bank, currently has a book value of ₹80 per share and an ROE of 25%, expected to persist for six years as it scales up its retail deposit and lending franchise, with a 10% payout maintained throughout this phase to comply with minimum distribution norms. Thereafter, ROE is expected to normalize to 16% with a stable payout ratio of 45%. The cost of equity is 15%. An analyst wants the two-stage justified P/B implied by this profile.

Required

1. Project the book value per share at the end of the six-year high-growth phase, accounting for the 10% payout.
2. Estimate the present value of the dividends paid during the high-growth phase, and the terminal-stage justified P/B and value at year six.
3. Estimate the implied two-stage justified P/B as of today.

Solution

A. Relevant Valuation Principle: The complete two-stage P/B has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal multiple applied to book value at the end of that phase. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — Growth rate during the high-growth phase

Formula: $g_{hg} = (1 - \text{payout}_{hg}) \times \text{ROE}_{hg}$

Substitution: $g_{hg} = (1 - 0.10) \times 0.25$

Result: **$g_{hg} = 22.5\%$**

Step 2 — Book value, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Book Value (BV_t)	$BV_0 \times (1+g_{hg})^t$	98.00	120.05	147.06	180.15	220.68	270.34
Dividend (D_t)	$\text{payout}_{hg} \times \text{ROE}_{hg} \times BV_t$	2.45	3.00	3.68	4.50	5.52	6.76
PV of Dividend	$D_t \div (1+K_e)^t$	2.13	2.27	2.42	2.58	2.74	2.92

Sum of PV of high-growth-phase dividends = ₹15.06

C. Stable Growth Period

Step 3 — Stable-stage growth rate

Formula: $g_{st} = (1 - \text{payout}_{st}) \times \text{ROE}_{st}$

Substitution: $g_{st} = (1 - 0.45) \times 0.16$

Result: **$g_{st} = 8.8\%$**

Step 4 — Stable-stage justified P/B

Formula: $\text{Justified P/B(stable)} = (\text{ROE}_{st} - g_{st}) / (K_e - g_{st})$

Substitution: $= (0.16 - 0.088) / (0.15 - 0.088)$

Result: **Justified P/B(stable) = 1.16x**

Step 5 — Value at the end of the high-growth period (Year 6)

Formula: $\text{Value}_6 = BV_6 \times \text{Justified P/B(stable)}$

Substitution: $= 270.34 \times 1.16$

Result: **Value₆ = ₹313.94**

Step 6 — Present value of Value6

Formula: $PV(\text{Value6}) = \text{Value6} \div (1+K_e)^6$

Substitution: $= 313.94 \div (1.15)^6$

Result: **PV(Value6) = ₹135.73**

D. Combining the Two Components*Step 7 — Total value today (P0)*

Formula: $P0 = \text{Sum of } PV(D_t) + PV(\text{Value6})$

Substitution: $= 15.06 + 135.73$

Result: **P0 = ₹150.78**

Step 8 — Implied two-stage justified P/B

Formula: $\text{Implied P/B} = P0 \div BV0$

Substitution: $= 150.78 \div 80$

Result: **Implied P/B = 1.88x**

E. Final Result: Book value grows to ₹270.34 per share by year 6; the high-growth phase separately contributes ₹15.06 of present value through its dividends; implied two-stage justified P/B as of today $\approx 1.88x$.

F. Interpretation: As in 42B, pairing each dividend with its own year's book value (rather than the smaller prior-year figure) raises every term in the interim-dividend stream, lifting the justified P/B from what a lagged-book-value convention would have produced (1.85x) to 1.88x.

G. Decision/Recommendation: The analyst should use 1.88x as the benchmark, built consistently on same-year book value throughout.

H. Teaching Insight: Read together, 42A–42C now show a clean progression: zero payout (42A) collapses the model to its terminal-only special case; any positive payout (42B, 42C) activates the interim-dividend term, computed consistently on each year's own book value — and the size of that term scales with both the payout rate and the number of high-growth years it is paid over.

Case 47A: Two-Stage P/S Valuation — Pharmacy Chain

Learning Objective: Estimate P/S when sales grow rapidly and payout changes after the high-growth period.

Case Situation

An organised pharmacy chain currently has sales per share of ₹100 and expects sales growth of 25% annually for the next five years as it expands into new cities. During this expansion, net margin is thin at 4% and payout is low at 10%, as most cash is reinvested. Thereafter, sales growth stabilises at 8%, with net margin improving to 6% and payout rising to 50%. The cost of equity is 14%. An analyst wants the two-stage justified P/S and a comparison with the current market multiple.

Required

1. Project sales per share and dividends per share for each of the five high-growth years.
2. Estimate the stable-stage terminal value and the resulting two-stage justified P/S.
3. Compare the result with the market multiple, assuming the stock currently trades at 1.2x sales.

Solution

A. Relevant Valuation Principle: The complete two-stage P/S has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal value based on improved margin and payout. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — Sales, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4	Year 5
Sales per Share (Sales _t)	$Sales_{t-1} \times (1+g_{hg})$	125.00	156.25	195.31	244.14	305.18
Dividend (D _t)	$Sales_t \times margin_{hg} \times payout_{hg}$	0.50	0.62	0.78	0.98	1.22
PV of Dividend	$D_t \div (1+Ke)^t$	0.44	0.48	0.53	0.58	0.63

Sum of PV of high-growth-phase dividends = ₹2.66

C. Stable Growth Period

Step 2 — Sales in the first stable year (Year 6)

Formula: $Sales_6 = Sales_5 \times (1+g_{stable})$

Substitution: $= 305.18 \times 1.08$

Result: **Sales₆ = ₹329.59**

Step 3 — Dividend in the first stable year

Formula: $D_6 = \text{Sales}_6 \times \text{margin_stable} \times \text{payout_stable}$

Substitution: $= 329.59 \times 0.06 \times 0.50$

Result: **D₆ = ₹9.89**

Step 4 — Terminal value at the end of Year 5

Formula: $TV_5 = D_6 \div (K_e - g_stable)$

Substitution: $= 9.89 \div (0.14 - 0.08)$

Result: **TV₅ = ₹164.79**

Step 5 — Present value of TV₅

Formula: $PV(TV_5) = TV_5 \div (1 + K_e)^5$

Substitution: $= 164.79 \div (1.14)^5$

Result: **PV(TV₅) = ₹85.59**

D. Combining the Two Components

Step 6 — Total value today (V₀)

Formula: $V_0 = \text{Sum of } PV(D_t) + PV(TV_5)$

Substitution: $= 2.66 + 85.59$

Result: **V₀ = ₹88.25**

Step 7 — Implied two-stage justified P/S

Formula: $\text{Implied P/S} = V_0 \div \text{Sales}_0$

Substitution: $= 88.25 \div 100$

Result: **Implied P/S = 0.88x**

E. Final Result: Implied two-stage justified P/S $\approx 0.88x$, against a market multiple of $1.2x$.

F. Interpretation: Despite the pharmacy chain's rapid 25% sales growth, the thin margins and low payout during the expansion phase mean very little value comes from the explicit high-growth dividends — almost all the implied value (about 97%) comes from the terminal value once margins and payout improve. The market's $1.2x$ multiple sits meaningfully above the $0.88x$ justified figure, suggesting the market expects either faster margin improvement, a longer growth runway, or lower risk than assumed here.

G. Decision/Recommendation: Given the market multiple exceeds the fundamentals-based justified P/S, the analyst should treat the stock as fully valued to slightly expensive at $1.2x$ sales, and should specifically probe the margin-improvement assumption before considering a Buy.

H. Teaching Insight: This case shows students that rapid revenue growth alone does not justify a high P/S multiple if margins remain thin during the growth phase — the terminal-stage margin assumption is doing almost all of the work in this valuation, similar to the terminal-value dominance seen in DDM and DCF cases.

Case 47B: Two-Stage P/S Valuation — Quick-Service Restaurant Chain

Learning Objective: Estimate P/S when sales grow very rapidly off a thin-margin, low-payout base, and recognize when a market multiple implies an unsustainable growth or margin story.

Case Situation

Zaika Quick Service Restaurants Ltd, an Indian QSR chain, currently has sales per share of ₹80 and expects aggressive sales growth of 30% annually for the next four years as it opens new outlets across tier-2 cities. During this expansion, net margin is thin at 3% and payout is minimal at 5%, since almost all cash is reinvested into store build-out. Thereafter, sales growth moderates to 7%, with net margin improving to 7% and payout rising to 40%. The cost of equity is 13%. An analyst wants the two-stage justified P/S and a comparison with the current market multiple.

Required

1. Project sales per share and dividends per share for each of the four high-growth years.
2. Estimate the stable-stage terminal value and the resulting two-stage justified P/S.
3. Compare the result with the market multiple, assuming the stock currently trades at 1.5x sales.

Solution

A. Relevant Valuation Principle: The complete two-stage P/S has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal value based on improved margin and payout. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — Sales, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4
Sales per Share (Sales _t)	$Sales_{(t-1)} \times (1+g_{hg})$	104.00	135.20	175.76	228.49
Dividend (D _t)	$Sales_t \times margin_{hg} \times payout_{hg}$	0.16	0.20	0.26	0.34

Item	Formula	Year 1	Year 2	Year 3	Year 4
PV of Dividend	$D_t \div (1+K_e)^t$	0.14	0.16	0.18	0.21

Sum of PV of high-growth-phase dividends = ₹0.69

C. Stable Growth Period

Step 2 — Sales in the first stable year (Year 5)

Formula: $Sales_5 = Sales_4 \times (1+g_{stable})$

Substitution: $= 228.49 \times 1.07$

Result: **Sales₅ = ₹244.48**

Step 3 — Dividend in the first stable year

Formula: $D_5 = Sales_5 \times margin_{stable} \times payout_{stable}$

Substitution: $= 244.48 \times 0.07 \times 0.40$

Result: **D₅ = ₹6.85**

Step 4 — Terminal value at the end of Year 4

Formula: $TV_4 = D_5 \div (K_e - g_{stable})$

Substitution: $= 6.85 \div (0.13 - 0.07)$

Result: **TV₄ = ₹114.09**

Step 5 — Present value of TV₄

Formula: $PV(TV_4) = TV_4 \div (1+K_e)^4$

Substitution: $= 114.09 \div (1.13)^4$

Result: **PV(TV₄) = ₹69.97**

D. Combining the Two Components

Step 6 — Total value today (V₀)

Formula: $V_0 = \text{Sum of } PV(D_t) + PV(TV_4)$

Substitution: $= 0.69 + 69.97$

Result: **V₀ = ₹70.66**

Step 7 — Implied two-stage justified P/S

Formula: $\text{Implied P/S} = V_0 \div Sales_0$

Substitution: $= 70.66 \div 80$

Result: **Implied P/S = 0.88x**

E. Final Result: Implied two-stage justified P/S $\approx 0.88x$, against a market multiple of $1.5x$.

F. Interpretation: Despite a punishing 30% sales growth rate, the terminal value contributes about 99% of total value — the explicit four-year growth phase, at a 3% margin and 5% payout, contributes almost nothing. The market's $1.5x$ multiple sits far above the $0.88x$ justified figure — a much wider gap than a typical growth story would suggest, implying the market is pricing in either a materially longer runway of 30% growth, a much faster margin ramp than modelled, or is simply chasing the growth headline without discounting it.

G. Decision/Recommendation: The gap here (roughly 70% above the justified multiple) is too large to attribute to reasonable optimism alone. The analyst should treat the stock as significantly overvalued at $1.5x$ sales and should specifically challenge whether store-level unit economics can plausibly improve margin from 3% to 7% as quickly as assumed.

H. Teaching Insight: This case is a deliberately more extreme version of the terminal-value-dominance lesson: even a very high headline growth rate is nearly worthless to value if it comes at a thin margin and minimal payout. Students should notice that the size of the market-justified gap matters — a 36% gap (as in a typical case) invites a "fully valued" conclusion, while a 70% gap like this one should prompt a harder challenge to the market's implicit assumptions, not just a mild caution.

Case 47C: Two-Stage P/S Valuation — Specialty Packaged Foods Company

Learning Objective: Estimate P/S for a moderate-growth company and recognize that a two-stage justified multiple can also reveal undervaluation, not just overvaluation.

Case Situation

Priyam Specialty Foods Ltd, an Indian branded packaged-foods company, currently has sales per share of ₹150 and expects steady sales growth of 20% annually for the next six years as its products gain shelf space in modern retail. During this expansion, net margin is 5% and payout is 15%. Thereafter, sales growth settles at 6%, with net margin improving to 8% and payout rising to 60%. The cost of equity is 12%. An analyst wants the two-stage justified P/S and a comparison with the current market multiple.

Required

1. Project sales per share and dividends per share for each of the six high-growth years.
2. Estimate the stable-stage terminal value and the resulting two-stage justified P/S.
3. Compare the result with the market multiple, assuming the stock currently trades at $1.0x$ sales.

Solution

A. Relevant Valuation Principle: The complete two-stage P/S has two components — the present value of dividends paid *during* the high-growth phase, and the present value of a stable-stage terminal value based on improved margin and payout. Each is worked out separately below, one step at a time, before the two are combined.

B. High-Growth Period

Step 1 — Sales, dividend, and present value of dividend, year by year

Item	Formula	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Sales per Share (Sales _t)	$Sales_{(t-1)} \times (1+g_{hg})$	180.00	216.00	259.20	311.04	373.25	447.90
Dividend (D _t)	$Sales_t \times margin_{hg} \times payout_{hg}$	1.35	1.62	1.94	2.33	2.80	3.36
PV of Dividend	$D_t \div (1+Ke)^t$	1.21	1.29	1.38	1.48	1.59	1.70

Sum of PV of high-growth-phase dividends = ₹8.65

C. Stable Growth Period

Step 2 — Sales in the first stable year (Year 7)

Formula: $Sales_7 = Sales_6 \times (1+g_{stable})$

Substitution: $= 447.90 \times 1.06$

Result: **Sales₇ = ₹474.77**

Step 3 — Dividend in the first stable year

Formula: $D_7 = Sales_7 \times margin_{stable} \times payout_{stable}$

Substitution: $= 474.77 \times 0.08 \times 0.60$

Result: **D₇ = ₹22.79**

Step 4 — Terminal value at the end of Year 6

Formula: $TV_6 = D_7 \div (Ke - g_{stable})$

Substitution: $= 22.79 \div (0.12 - 0.06)$

Result: **TV₆ = ₹379.82**

Step 5 — Present value of TV₆

Formula: $PV(TV_6) = TV_6 \div (1+Ke)^6$

Substitution: $= 379.82 \div (1.12)^6$

Result: **PV(TV₆) = ₹192.43**

D. Combining the Two Components

Step 6 — Total value today (V0)

Formula: $V0 = \text{Sum of PV}(D_t) + \text{PV}(TV6)$

Substitution: $= 8.65 + 192.43$

Result: **V0 = ₹201.08**

Step 7 — Implied two-stage justified P/S

Formula: $\text{Implied P/S} = V0 \div \text{Sales}_0$

Substitution: $= 201.08 \div 150$

Result: **Implied P/S = 1.34x**

E. Final Result: Implied two-stage justified P/S $\approx 1.34x$, against a market multiple of $1.0x$.

F. Interpretation: Here the relationship reverses: the market's $1.0x$ multiple sits below the fundamentals-based justified figure of $1.34x$. With a moderate 5% margin, a meaningful 15% payout even during the growth phase, and a credible improvement to an 8% margin thereafter, the terminal value (about 96% of total value) is well supported by realistic assumptions — yet the market is pricing the stock as if growth or margin expansion will disappoint.

G. Decision/Recommendation: Unlike Cases 47A and 47B, this case supports a Buy: the stock appears undervalued by roughly 25% relative to its justified multiple, provided the analyst is confident the 8% stable-stage margin is achievable — a materially smaller assumption stretch than the margin jump required in Case 47B.

H. Teaching Insight: Place this case immediately after 47A and 47B to make the point explicit: the two-stage P/S method is not a one-directional "growth stocks are overvalued" tool — it is equally capable of flagging undervaluation. What changes the verdict is not the growth rate itself, but the credibility of the margin and payout assumptions supporting the terminal value in each case.